Computer System Architecture

Lab 2 - Digital Logic Review

September 2026

Question 1

We use 8-bit two's complement system.

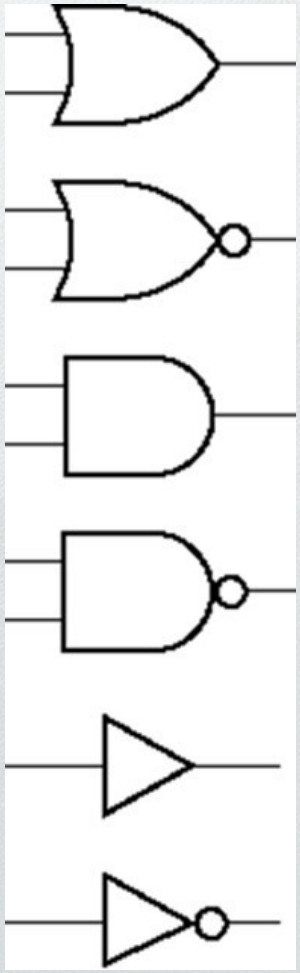
1. Compute $73_{10} + 58_{10}$ in 8-bit two's complement binary.
2. Check whether signed integer overflow occurs. Give your reasoning.
3. What is the interpreted decimal result from the 8-bit binary output?

Question 2

Construct a truth table for the Boolean expression:

$$X = (A + B)(B + C)(C + A)$$

Logic Gates



OR

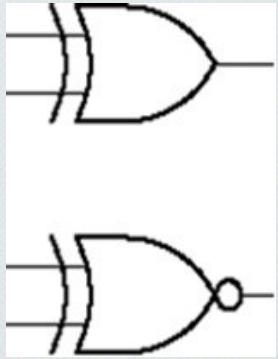
NOR

AND

NAND

buffer

NOT (inverter)



XOR

XNOR

Question 3

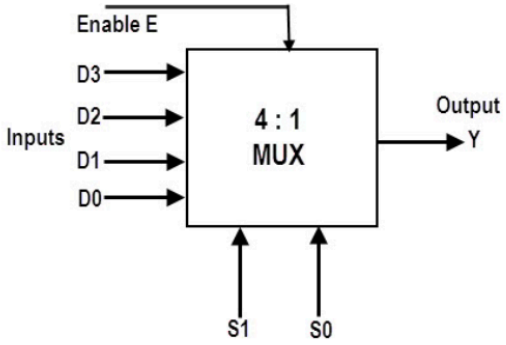
Implement the XOR gate using NAND gates only.



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Question 4

Design the logic diagram of a full adder with inputs A, B, C_{in} and outputs Σ, C_{out}, by using 4-to-1 multiplexers (ref. Figure 3) and NOT gates.

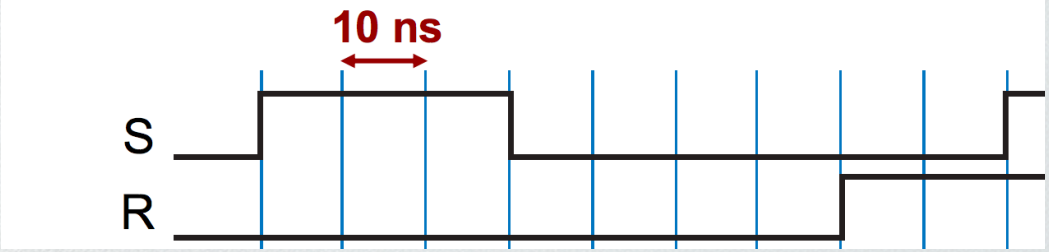


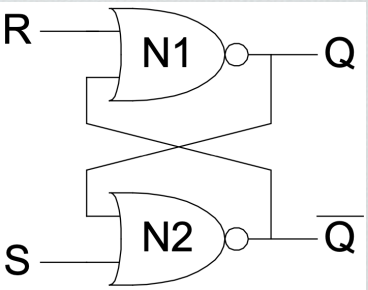
- **Truth Table for Full Adder -**

Inputs			Outputs	
A	B	C-In	Sum	C-Out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Question 5

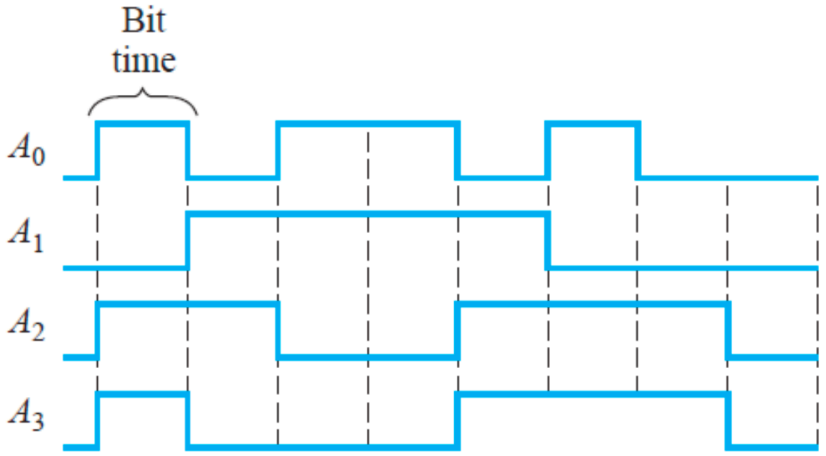
Draw the outputs of an SR latch for the input waveforms shown below. Assume that the propagation delay of a NOR gate is 10ns.

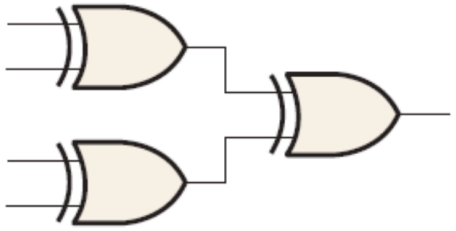




Question 6

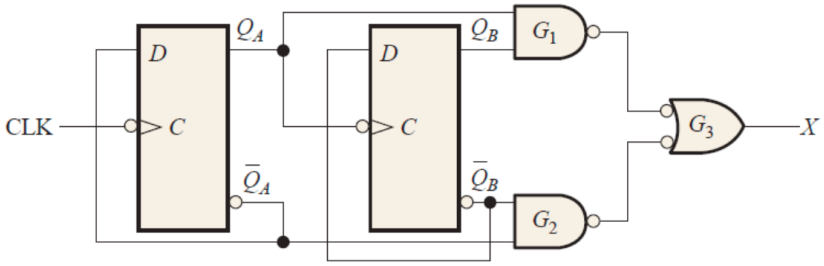
The waveforms in Figure 6–82 are applied to the 4-bit parity logic. Determine the output waveform in proper relation to the inputs. For how many bit times does even parity occur, and how is it indicated? The timing diagram includes eight bit times.



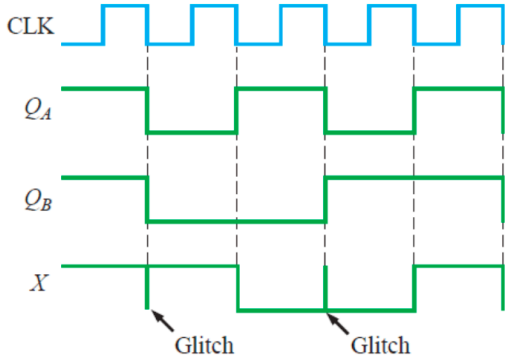


Question 7

The flip-flop circuit in Figure 7–95(a) is used to generate a binary count sequence. The gates form a decoder that is supposed to produce a HIGH when a binary zero or a binary three state occurs (00 or 11). When you check the Q_A and Q_B outputs, you get the display shown in part (b), which reveals glitches on the decoder output (X) in addition to the correct pulses. What is causing these glitches, and how can you eliminate them?



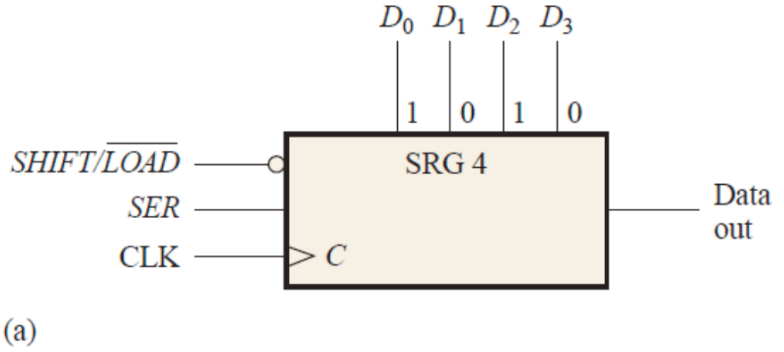
(a)



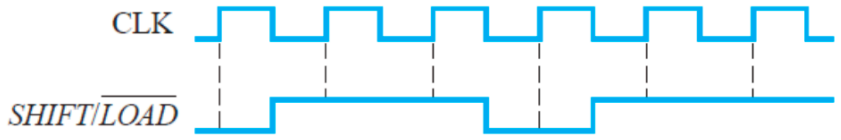
(b)

Question 8

The shift register in Figure 8–54(a) has $SHIFT/\overline{LOAD}$ and CLK inputs as shown in part (b). The serial data input (SER) is a 0. The parallel data inputs are $D_0 = 1$, $D_1 = 0$, $D_2 = 1$, and $D_3 = 0$ as shown. Develop the data-output waveform in relation to the inputs.



(a)



(b)

FIGURE 8–54

Question 9

Determine the Q outputs of a 74HC194 with the inputs shown in Figure 8–59. Inputs D_0, D_1, D_2, and D_3 are all HIGH.

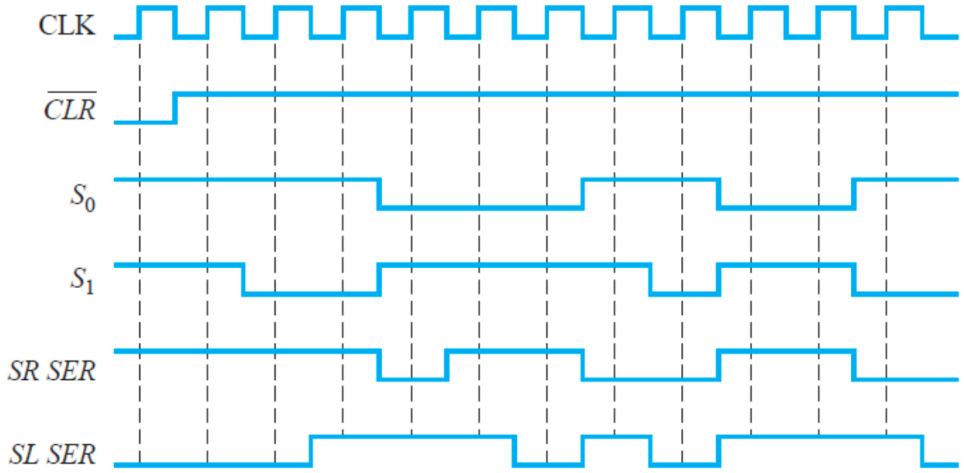


FIGURE 8–59

Question 10

For the ripple counter in Figure 9–66, show the complete timing diagram for sixteen clock pulses. Show the clock, Q_0, Q_1, and Q_2 waveforms.

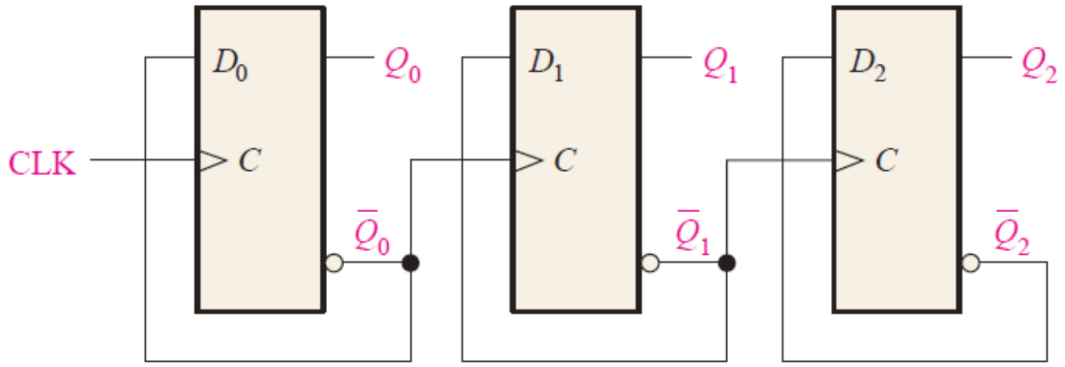


FIGURE 9–66